

Few-Step Autoregressive Latent Flow Video Models: Drift, Contraction, Sampler Robustness and Synthetic-Rollout Bias

Executive summary

No paper located through **2 August 2026** reports the complete controlled dissociation in your experiments: **distillation-induced directional colour drift, same-weights few-step hue-neutral blur/contrast loss**, and **fixed-noise endpoint-ladder structural melting that is mitigated by fresh-noise endpoint re-noising**. Several recent papers nevertheless report each component separately, often in nearly identical Wan2.1, three-latent-frame, four-step autoregressive settings.

The strongest direct video evidence is that long autoregressive rollouts develop **DC/low-frequency spectral drift, colour change, fading and sharpness loss**; low-frequency anchoring and pathwise corrections measurably reduce them. Image-flow work further shows that global colour is represented by low-dimensional directions in the spatially averaged VAE latent, supporting a decoder-visible **channel-mean vector** interpretation rather than a one-channel-one-colour mapping. Your failure of a scalar gain, combined with AdaIN success, therefore indicates anisotropic mean/covariance error, although mean-only versus variance-only ablations are still needed to identify the causal term.

Your stochastic sampler has an unusually close published analogue: Stable Audio 3's **ping-pong sampling** repeatedly predicts x_0 , re-noises it to a lower absolute level with fresh noise, and substantially outperforms one-shot or deterministic propagation. Restart and ODE-versus-SDE analyses support the broader mechanism—stochastic re-projection can suppress temporally correlated field error—but do not directly cover your monotone ladder or fixed- ϵ_0 comparator.

For training, the closest video literature increasingly favours **student-on-policy states, cache-aware mixed-step training, self-resampled histories, transition/flow-map targets, and real-data or strong-reference anchors**. Pure endpoint MSE on teacher-generated histories combines conditional averaging, inherited teacher bias and off-policy cache exposure.

The best unified interpretation is:

observed failure = directional learned DC bias + few-step centred-energy contraction + sampler-dependent resampling error
all multiplied by autoregressive cache feedback.

Scope, terminology and evidence map

Exact relationship to your system

Your stochastic update is

$$\hat{x}_{0,k} = F_{\theta,k}(x_{\sigma_k}, h_n), \quad x_{\sigma_{k+1}} = (1 - \sigma_{k+1})\hat{x}_{0,k} + \sigma_{k+1}\epsilon_{k+1},$$

with independent $\epsilon_{k+1} \sim \mathcal{N}(0, I)$. The deterministic comparator uses the same endpoint ladder but sets every $\epsilon_{k+1} = \epsilon_0$. This distinction matters:

- it is **not** an Euler–Maruyama discretisation of a reverse SDE;
- it is **not** EDM churn, which perturbs an existing noisy state by temporarily increasing its noise level;
- it is **not** a standard probability-flow ODE solver;
- and the fixed-noise version is **not** implied-noise inversion.

It is best described as a **monotonically descending endpoint re-projection ladder**, with noise correlation $\rho = 0$ for fresh noise and $\rho = 1$ for fixed initial noise.

The closest complete architectural precedent is [Self Forcing](#): Wan2.1-1.3B, a causal 3D VAE, three latent frames per autoregressive chunk, four sampling steps, shifted timesteps, cached context, and repeated x_0 prediction followed by fresh-noise re-noising. Its training, however, uses self-rollouts and distribution-level objectives rather than MSE against stored teacher endpoints or transitions. ¹

Comparison of the principal papers

“Direct” below means evidence in autoregressive video; “analogue” means image, audio or non-AR evidence with a matching mechanism.

Paper	Domain and model family	Sampler and distillation	Colour or DC drift	Contraction or blur	Structural amplification	Mitigation and quantitative results
FreqForcing	Long AR video; Wan2.1/Self-Forcing; latent flow model	Four-step AR generation; training-free spectral anchoring	Direct: colour drift and DC/low-frequency energy drift	Direct: fading and energy collapse	Long-rollout collapse/motion stagnation	At 60 s, dynamic degree 33.29 → 59.58, overall consistency 18.81 → 20.94, imaging quality 66.90 → 69.52
Pathwise Test-Time Correction	AR video; Wan2.1, CausVid and Self Forcing	Corrects intermediate path states, then re-noises and continues	Direct: first/last colour-histogram drift	Some appearance/flicker evidence	Cache/path accumulation	Self Forcing colour histogram L1 1.028 → 0.644, correlation 0.479 → 0.710; temporal LPIPS 0.205 → 0.176
Why Do DiT Editors Drift?	Repeated image editing; FLUX.2 and SD3-type latent models	Repeated latent transitions; low-frequency affine alignment	Direct analogue: per-channel mean identified with DC offset	Low-frequency energy/structure degradation	Repeated-transition drift	Tenth-round FLUX. example: LPIPS 0.36 → 0.17, SSIM 0.81 → 0.93 under low-frequency alignment

Paper	Domain and model family	Sampler and distillation	Colour or DC drift	Contraction or blur	Structural amplification	Mitigation and quantitative results
The Latent Color Subspace	FLUX.1 flow-matching images; VAE latent	Closed-form latent subspace intervention	Direct analogue: HSL-like colour directions in global latent statistics	Not a rollout-contraction study	No	GenEval colour accuracy 9% $\rightarrow$ 73%; plain-image $\Delta E_{00} = 9$ versus 22 for prompt control
Stable Audio 3	Latent audio flow model	Fresh-noise x_0 -predict/re-noise “ping-pong” sampling	No	One-step quality loss	Explicitly motivates correction of displaced deterministic trajectories	Medium model FAD 0.258 $\rightarrow$ 0.107, CLAP 0.355 $\rightarrow$ 0.390, one to eight ping-pong steps; eight steps beat the 50-step base FAD 0.143
Restart Sampling	Image generative processes	Upward noise restart followed by downward ODE transport	No	Numerical/model error analysis	Theory analogue: deterministic error accumulation and stochastic contraction	Outperforms ODE and SDE baselines over multiple NFE budgets; reported CIFAR-10 FID about 1.90 at 43 NFE versus 1.97 for a higher-NFE deterministic baseline
Self Forcing	AR Wan2.1 video flow model	Four-step fresh-noise endpoint ladder; self-rollout distribution matching	Reports saturation drift in prior AR methods	Long-horizon degradation	Cache exposure addressed by self-rollouts	Approximately 17 FPS and sub-second initial latency; improves long-video quality over teacher-forced/CausVid settings
Salt	Wan2.1 and AR Self-Forcing/Causal-Forcing video	SC-DMD; mixed $K \in \{2, 4, 8\}$ cache rollouts; semigroup-defect penalty	Over-exposure reported as compositional drift	Direct: trajectory regression becomes over-smoothed with weak motion	Direct: locally trained updates drift when composed	Four-NFE AR total 84.62 $\rightarrow$ 85.08; at two NFE, reference alignment raises semantic 80.65 $\rightarrow$ 81.49

Paper	Domain and model family	Sampler and distillation	Colour or DC drift	Contraction or blur	Structural amplification	Mitigation and quantitative results
OPSD-V	Four-step AR Wan2.1 video	Student-on-policy cache; dense-step teacher at student states; velocity matching	Not specifically isolated	Reports x_0 -space smoothing/geometric degradation	Direct cache-distribution mismatch	LongLive quality 0.8138 $\rightarrow$ 0.8242 dynamics 0.5012 $\rightarrow$ 0.5890 Self Forcing quality 0.8259 $\rightarrow$ 0.8389
Self-Resampling / Resampling Forcing	AR Wan2.1 flow video, three latent frames/chunk	Online model resamples degraded histories; clean real target	Reports progressive colour, texture and sharpness degradation in baselines	Direct	Direct: teacher-forcing exposure bias	AR self-resampling temporal/visual/text scores 90.46/64.25/25.2 versus 87.15/61.90/21.4 for noise augmentation
On-Policy Adversarial Flow Distillation	Two causal AR video families	Student rollouts; black-box teacher clips; advantage-weighted forward FM	No isolated DC test	SFT can worsen quality/motion	Off-policy SFT explicitly diagnosed	Self-Forcing motion quality 0.341 $\rightarrow$ 0.605, physical consistency 4.04 $\rightarrow$ 4.20; SFT falls to 0.296/3.72
Transition Matching Distillation	Wan2.1-1.3B/14B video flow models	Wide noise-level probability transitions implemented with conditional inner flows	No	Designed to preserve details across large transitions	Transition composition addressed architecturally	14B reaches VBench 84.24 at effective NFE 1.38; up to roughly 70 $\times$ acceleration
AnyFlow	Bidirectional and causal video flow models, 1.3B-14B	On-policy arbitrary-time flow-map learning $z_t \rightarrow z_r$	No	Endpoint-consistency models can worsen when given more steps	Exposure and discretisation errors treated jointly	Matches or exceeds fixed few-step consistency counterparts while improving with added test-time steps

Paper	Domain and model family	Sampler and distillation	Colour or DC drift	Contraction or blur	Structural amplification	Mitigation and quantitative results
CausVid	Four-step causal video student from bidirectional teacher	ODE initialisation plus distribution matching	Later papers report saturation/appearance drift	Some long-rollout degradation	Teacher/student causality mismatch	VBench-Long total 84.27; ODE initialisation plus bidirectional teacher improves short-clip temporal frame/text scores to 94.7/64.4/30.1
ODE versus SDE	Analytical and image diffusion analogue	Probability-flow ODE versus stochastic process under field error	No	No video-specific blur	Theory: early/mid-course field error suppressed by stochasticity; late error may favour ODE	Mathematical pulse-error analysis plus Gaussian, mixture, MNIST and CIFAR validation

The table reveals an important gap: published work usually compares training recipes, sampler families or low-frequency corrections separately. None performs your controlled decomposition of **weights**, **NFE** and **noise-correlation regime** while tracing the same 16-dimensional latent statistics.

Chromatic drift, latent DC motion and few-step contraction

What has actually been reported in video

[FreqForcing](#) is the nearest direct report. It traces latent- and pixel-space spectral energy during long Self-Forcing rollouts and identifies rapid changes in the DC and low-frequency bands alongside colour drift, fading, motion stagnation and eventual collapse. Enlarging the attention-sink context slows but does not remove the drift, whereas explicit spectral self-anchoring stabilises low-frequency energy. On 60-second rollouts, dynamic degree rises from 33.29 to 59.58, imaging quality from 66.90 to 69.52, aesthetic quality from 56.77 to 60.91 and overall consistency from 18.81 to 20.94. At 120 seconds, dynamic degree rises from 27.42 to 58.97. Stronger anchoring increases repetition, demonstrating that exact low-frequency preservation is not cost-free. ²

[Pathwise Test-Time Correction](#) separately quantifies colour-histogram drift in distilled Wan2.1 AR models. Between the initial and final portions of 30-second Self-Forcing videos, its correction reduces HSV-histogram L1 distance from 1.028 to 0.644 and raises histogram correlation from 0.479 to 0.710; for CausVid, L1 drops from 1.047 to 0.607 and correlation rises from 0.451 to 0.778. Correcting selected intermediate path states performs better than a single final-output correction, including temporal LPIPS 0.205 → 0.176 in a reported Self-Forcing comparison. ³

[Self-Resampling](#) explicitly describes strict AR baselines such as Pyramid Flow, CausVid and Self Forcing as suffering progressive degradation of **colour, texture and sharpness**. Its evidence does not separate those observables into different causal factors, but it shows that exposing the model to autoregressively resampled imperfect histories is more effective than generic Gaussian history

augmentation: temporal/visual/text scores are 90.46/64.25/25.26 for autoregressive self-resampling, 88.01/62.51/24.51 for parallel resampling and 87.15/61.90/21.44 for additive noise. 4

Thus, the literature clearly supports the existence of colour drift, low-frequency drift, fading and sharpness loss. It does **not** yet support the sharper statement that colour drift is trained in while hue-neutral blur is caused by low NFE. In published systems, the distilled weights, self-generated cache, training distribution and sampler are usually changed together.

Why your training-versus-sampling dissociation is unusually strong

Your same-weights NFE curve supplies precisely the control missing from those papers:

$$r_{\text{std}}^{(4)} \approx 0.917, \quad r_{\text{std}}^{(20)} \approx 0.983, \quad r_{\text{std}}^{(48)} \approx 0.987$$

per chunk for the base weights, while the trained four-step student retains approximately 0.956 per chunk but acquires an asymmetric colour cast.

That pattern argues for at least two distinct latent effects:

$$\begin{aligned} z_n &= \mu_n + \tilde{z}_n, & \mathbb{E}[\tilde{z}_n] &= 0, \\ \mu_{n+1} &= A_\mu \mu_n + b, & \Sigma_{n+1} &= A_\Sigma \Sigma_n A_\Sigma^\top + Q. \end{aligned}$$

The hue-neutral base-model degradation is naturally associated with contraction of the centred component $\tilde{z}_n$, reflected by falling channel standard deviations, luma contrast and fine-scale detail. The student’s colour cast is more naturally associated with the mean recurrence through b or anisotropic modes of A_μ . A single latent “quality” scalar cannot explain a student that retains more overall standard deviation than the four-step base yet develops greater chromatic asymmetry.

[Salt](#) provides an adjacent objective-level explanation for blur. It reports that trajectory-regression or consistency-style supervision becomes conservative in video because multiple plausible motion continuations are forced towards a point target, yielding over-smoothed appearance and weak motion. Conversely, distribution matching can remain sharp but has a “compositionality deficit”: independently trained denoising updates need not compose coherently, so more updates can produce over-exposure, semantic loss or drift. Its semigroup-defect regulariser and cache-aware mixed-step training improve a four-NFE AR total score from 84.62 to 85.08; at two NFE, high-step reference alignment raises semantic score from 80.65 to 81.49. 5

This is relevant to both of your targets. Regressing the final teacher sample from every rung asks high- σ states to identify a specific clean continuation despite considerable conditional ambiguity. Your refined $x_k \rightarrow x_{k+1}$ target localises the required transition and is therefore better posed, but it remains a pointwise MSE target on a single teacher realisation. It can still learn conditional averages in state directions not sufficiently identified by (x_k, h_n) .

Colour as a direction in VAE channel-mean space

The most direct geometric evidence comes from [The Latent Color Subspace](#). In FLUX.1’s flow-matching VAE latent, a low-dimensional subspace of global latent statistics has an HSL-like geometry: two directions represent circular hue structure, while additional directions encode saturation and lightness. Closed-form interventions in this subspace raise GenEval colour accuracy from 9% to 73% without adding a colour word to the prompt, close to the 79% attained by explicit prompting. On plain-colour

tests, the intervention reaches $\Delta E_{00} = 9$, hue error 11° and lightness error 3%, versus 22, 38° and 12% for prompt-only control. It also preserves structure better than changing the prompt, with global-intervention IoU 0.88 versus 0.60 and DINOv2 distance 0.23 versus 0.36. ⁶

This supports the following local decoder model for a spatially constant latent perturbation:

$$\Delta \bar{y}_{\text{RGB}} \approx J_{\text{DC}}(z) \Delta \mu_z, \quad J_{\text{DC}} \in \mathbb{R}^{3 \times 16},$$

where μ_z is the 16-channel mean and $\bar{y}_{\text{RGB}}$ is mean decoded colour. Since J_{DC} has at most rank three locally, only a low-dimensional set of directions in the 16-dimensional mean vector can strongly affect global RGB colour. It is therefore generally wrong to seek a stable “latent channel 7 equals green” correspondence. Colour is more plausibly encoded by **linear or mildly nonlinear combinations of channels**.

[Why Do DiT Editors Drift?](#) supplies the complementary repeated-transition evidence. It decomposes repeated editing drift by spatial frequency, identifies each channel’s spatial mean as its DC coefficient, and treats channel standard deviation as centred energy. Its VAE-LFA method aligns channel-wise mean and standard deviation of the low-pass latent to an exponentially averaged anchor while retaining the high-frequency residual. In reported tenth-round FLUX.2 no-op editing examples, LPIPS improves from 0.36 to 0.17 and SSIM from 0.81 to 0.93; low-frequency-only alignment generally outperforms high-frequency-only correction. ⁷

Together, these papers strongly support **directional DC-subspace motion** as a colour-drift mechanism. They do not demonstrate it in Wan’s 16-channel 3D VAE, so the decoder map must be measured rather than assumed transferable.

What AdaIN success and scalar-gain failure establish

A global gain implements

$$z' = gz, \quad \mu' = g\mu, \quad \Sigma' = g^2\Sigma.$$

It only moves statistics radially around the origin. A per-channel affine correction implements

$$z'_c = \mu_c^{\text{ref}} + \frac{s_c^{\text{ref}}}{s_c}(z_c - \mu_c),$$

and can undo a channel-dependent translation plus diagonal rescaling. Your intervention therefore establishes that the failure is not a uniform latent contraction.

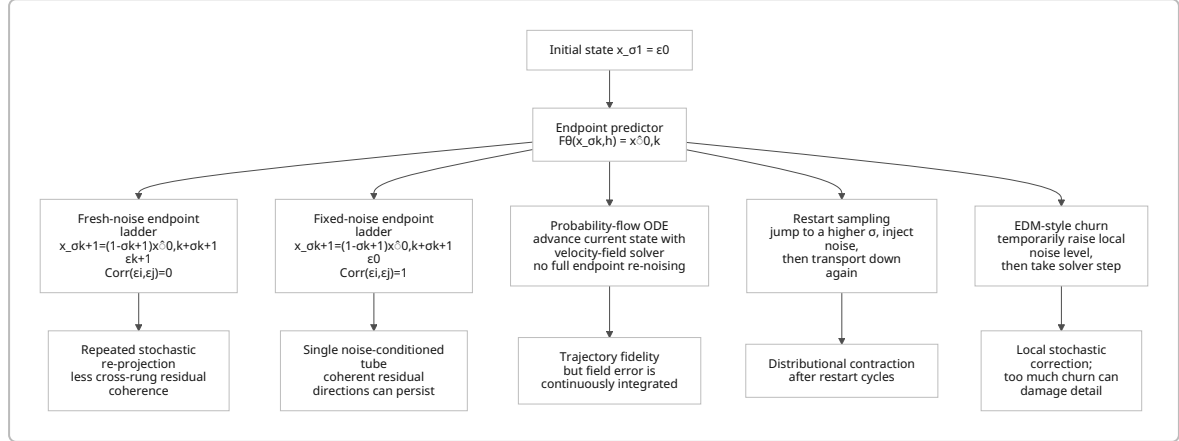
It does **not**, by itself, prove that the cast is caused solely by channel means. Full AdaIN changes both μ_c and s_c , and a nonlinear decoder can convert anisotropic variance changes into colour changes. The decisive decomposition is:

$$\begin{aligned} \text{mean-only: } z'_c &= z_c - \mu_c + \mu_c^{\text{ref}}, \\ \text{std-only: } z'_c &= \mu_c + \frac{s_c^{\text{ref}}}{s_c}(z_c - \mu_c), \\ \text{low-pass mean-only: } z' &= z - H_{\text{LP}}(\mu - \mu^{\text{ref}}), \\ \text{decoder-subspace: } \mu' &= \mu - P_{\text{chroma}}(\mu - \mu^{\text{ref}}). \end{aligned}$$

The literature predicts that mean-only, especially low-pass or chroma-subspace mean correction, should remove most of the cast; standard-deviation correction should recover contrast or texture but have a smaller effect on hue. ⁸

Fresh-noise re-projection, fixed-noise paths and structural melting

The sampler taxonomy



The $\rho = 0$ and $\rho = 1$ branches are your actual comparison. The other three are adjacent evidence, not algorithmic equivalents.

Exact fresh-noise analogue: Stable Audio 3

[Stable Audio 3](#) calls the same operation **ping-pong sampling**. At each stage it estimates the clean endpoint, re-noises that estimate to a lower absolute noise level using a new Gaussian draw, and predicts again. The authors contrast this with deterministic integration: once an inaccurate deterministic update moves a sample onto the wrong path, subsequent evaluations continue from that displaced state; re-noising the current endpoint estimate gives the model another opportunity to project towards the conditional data distribution. ⁹

For its medium music model, increasing post-trained ping-pong sampling from one to eight endpoint evaluations improves FAD from 0.258 to 0.107 and CLAP from 0.355 to 0.390. The eight-step model's FAD also beats the 50-step base model's 0.143. The large model improves from FAD 0.273 and CLAP 0.331 at one step to 0.101 and 0.393 at eight steps. This is unusually strong empirical evidence that repeated fresh-noise endpoint re-projection can correct errors rather than merely inject diversity. ¹⁰

It remains an audio result, and it compares fresh-noise ping-pong primarily with one-step generation and ordinary deterministic integration—not with your otherwise identical fixed- ϵ_0 ping-pong ladder. I found no published paper that makes that exact controlled comparison.

Why fixed noise can preserve metrics yet melt structurally

Let the ideal endpoint map at rung k be F_k^* , with learned residual

$$F_k(x, h) = F_k^*(x, h) + e_k(x, h).$$

Linearising consecutive fixed-noise updates around an ideal path gives

$$\delta x_{k+1} \approx (1 - \sigma_{k+1}) [J_{x,k} \delta x_k + J_{h,k} \delta h_n + e_k].$$

The same ϵ_0 keeps every rung on a highly correlated noise-conditioned tube. Consequently, errors associated with a particular object boundary, pose, depth arrangement or cached-attention configuration can have similar signs at successive rungs. The accumulated structural component contains products such as

$$\delta x_K \approx \sum_{j < K} \left[\prod_{\ell=j+1}^K (1 - \sigma_\ell) J_{x,\ell} \right] e_j.$$

Even if every one-step endpoint estimate is reasonably accurate, an unstable singular direction of the composed Jacobian can turn a coherent error into gradual geometric collapse.

Fresh noise changes the conditional input direction at each rung:

$$e_k = e_k(\epsilon_k, h_n), \quad \text{Cov}(e_i, e_j) \text{ is typically reduced when } \epsilon_i \perp \epsilon_j.$$

This does not literally average several final samples, because only one chain is retained. The more precise interpretation is **decorrelation plus repeated stochastic re-projection**. State-dependent errors that are odd, noise-direction-specific or locally inconsistent are less likely to remain aligned across all rungs. Systematic bias, however, survives:

$$\mathbb{E}[e_k] = b_k \neq 0 \implies \text{fresh zero-mean noise does not remove } b_k.$$

That explains why fresh noise could suppress melting while leaving a trained-in chromatic drift.

What restart and ODE-SDE theory add

[Restart Sampling](#) analyses a different algorithm: after transporting downwards, it adds substantial noise to jump back to a higher level and transports down again. Its central result is that the noisy restart followed by denoising can contract distributional error, balancing newly introduced solver error against correction of previously accumulated model error. It consistently outperforms tested deterministic and stochastic baselines over several datasets and function-evaluation budgets. ¹¹

The NeurIPS study [Exploring the Optimal Choice for Generative Processes: ODE versus SDE](#) gives a useful error-location qualification. Under an imperfect learned field, stochasticity can exponentially suppress errors introduced early or in the middle of generation, whereas errors concentrated very near the clean endpoint can favour deterministic ODE transport because insufficient time remains for stochastic correction. ¹²

Translated into your endpoint-ladder setting, the implication is not “stochastic is always better”. It is:

- fresh re-noising should help when the dominant error is generated at high or intermediate σ , is strongly state-dependent, and becomes harmful mainly through coherent propagation;
- deterministic or highly correlated noise should help when late-rung estimates are already accurate and exact seed-conditioned path fidelity matters;
- excessive late stochasticity can resample action-relevant or identity-relevant details after they have become well determined.

This predicts that a hybrid correlation schedule—fresh at $\sigma = 1$ and 0.625, partially correlated at 0.357, nearly fixed at 0.208—may occupy a better action-consistency/structural-robustness frontier than either extreme.

Why pointwise and action metrics can prefer fixed noise

The fixed- ϵ_0 chain preserves a coherent correspondence between the initial stochastic degrees of freedom and every later endpoint estimate. That can improve:

- exact latent or pixel correspondence;
- deterministic seed reproducibility;
- action-conditioned displacement measured against a particular target;
- short-horizon optical-flow alignment.

Fresh re-noising marginalises some unresolved detail by repeatedly drawing from the model's conditional neighbourhood. It can look more plausible while moving further from a particular target trajectory. Thus the relevant distinction is:

fixed-noise transport: better path identity, potentially worse coherent error;
fresh-noise re-projection: better local manifold recovery, potentially worse pointwise identity.

The video literature increasingly recognises the same fidelity-versus-robustness trade-off in history perturbation. Self-Resampling finds weak perturbations preserve history but continue accumulating error, whereas overly strong resampling reduces semantic consistency and causes content drift; intermediate corruption performs best. ⁴

Distillation on generated autoregressive rollouts

Why your setting is off-policy even though the teacher paths are internally exact

For training chunk n , your student sees

$$(x_{\sigma_k, n}^T, h_n^T)$$

and learns a teacher endpoint or transition. At deployment, it sees

$$(x_{\sigma_k, n}^S, h_n^S).$$

The stored target can be algebraically exact for the teacher transition while still being statistically off-policy for the student. The mismatch enters through both the noisy state and the cached history:

$$p_{\text{train}}(x_\sigma, h) = p_T(x_\sigma, h), \quad p_{\text{test}}(x_\sigma, h) = p_S(x_\sigma, h).$$

Because h_n contains committed chunks and their cached features, a small student-specific colour or geometry error changes every later attention query/key interaction. Teacher forcing on teacher-generated rather than real histories removes one real-versus-generated mismatch, but leaves the more relevant **teacher-versus-student occupancy mismatch**.

[On-Policy Adversarial Flow Distillation](#) demonstrates this directly. It finds that supervised fine-tuning on teacher-generated videos trains under teacher prefixes and does not improve monotonically with more optimisation; VBench and physics scores can decline. On Self-Forcing, SFT lowers motion quality from

0.341 to 0.296 and physical consistency from 4.04 to 3.72. Its on-policy AFD instead generates student rollouts, compares them with teacher videos and turns the resulting advantage into forward-process velocity-field supervision, reaching 0.605 motion quality and 4.20 physical consistency. ¹³

Closest direct remedies

[OPSD-V](#) uses the exact four-step AR inference loop to produce student cache states. The teacher is queried at the student’s noisy state, but older context is replaced with clean real-video history while the immediately preceding student chunk is retained. This separates learning to correct student-visited states from blindly imitating a fully synthetic history. On LongLive, quality rises from 0.8138 to 0.8242 and dynamic degree from 0.5012 to 0.5890; on Self Forcing, quality rises from 0.8259 to 0.8389 and dynamics from 0.6218 to 0.6570. ¹⁴

[Self-Resampling](#) takes a teacher-free route. It partially corrupts real frames and lets the current online model complete their denoising autoregressively, creating history errors matched to the model’s current failure distribution. Those degraded histories condition a clean real-video flow-matching target. Autoregressive resampling outperforms both additive Gaussian corruption and independent per-frame resampling, suggesting that generic noise is an inadequate proxy for structured AR error. ⁴

[Salt](#) treats cache quality as an explicit condition. It mixes two-, four- and eight-step rollouts during training and aligns weak low-step cache features to stronger high-step references. This is especially relevant to your same-weights NFE result: a model trained only on one cache-quality regime can become brittle when its actual stepwise contraction differs from the training teacher’s. ⁵

[Self Forcing](#) also addresses occupancy mismatch by rolling out the student with the actual KV cache before applying a holistic distribution objective, rather than training only on externally generated fixed histories. It is one reason the paper is a closer conceptual match to your exposure problem than ordinary offline progressive distillation, despite having a different loss. ¹

Endpoint targets versus transition or flow-map targets

Your refined target

$$\hat{x}_0 = x_k - \sigma_k \frac{x_k - x_{k+1}}{\sigma_k - \sigma_{k+1}}$$

is algebraically a next-state transition constraint under the chosen linear interpolation and fixed path noise. It is more local than asking every rung to regress the teacher’s final clean sample. Recent video work increasingly supports that direction.

[Transition Matching Distillation](#) distils broad teacher trajectory segments into learned probability transitions between separated noise levels. Each outer transition uses a semantic backbone plus a lightweight conditional flow head for inner refinement; the target is distributional transition matching rather than pointwise regression to one teacher endpoint. Its Wan2.1-14B model reaches VBench 84.24 at an effective NFE of 1.38 and reports up to roughly 70× acceleration. ¹⁵

[AnyFlow](#) argues more explicitly that fixed endpoint consistency $z_t \rightarrow z_0$ replaces the teacher’s transport trajectory with a different consistency-sampling trajectory; quality can then worsen when more test-time steps are supplied. It instead learns arbitrary interval maps $z_t \rightarrow z_r$ and performs backward simulation on student-visited trajectories. The method targets both few-step discretisation error and

causal exposure bias, and—unlike many fixed-step consistency models—continues to improve as the test-time step budget grows. ¹⁶

These papers do not imply that your algebraic target is wrong. They imply that a **pointwise MSE transition along one fixed teacher noise realisation** may still be too narrow. Distributional transition matching, arbitrary-interval flow maps or student-on-policy next-state targets avoid forcing all admissible continuations towards the one stored teacher realisation.

Relation to self-consuming generative loops

The Nature paper [AI Models Collapse When Trained on Recursively Generated Data](#) defines model collapse across successive model generations trained on synthetic outputs. It finds that rare modes disappear first and late collapse can strongly reduce variance; access to the original human data distribution is crucial. ¹⁷

Your setup is not full multi-generation model collapse: one student is distilled from one teacher dataset. The closer labels are:

- **synthetic-target bias inheritance;**
- **offline imitation under teacher occupancy;**
- **closed-loop exposure bias;**
- and **within-rollout recursive amplification.**

Nevertheless, the same error taxonomy applies. Finite synthetic coverage loses parts of the real continuation distribution; limited student capacity introduces approximation bias; MSE adds objective-induced conditional averaging; and autoregressive reuse amplifies all three.

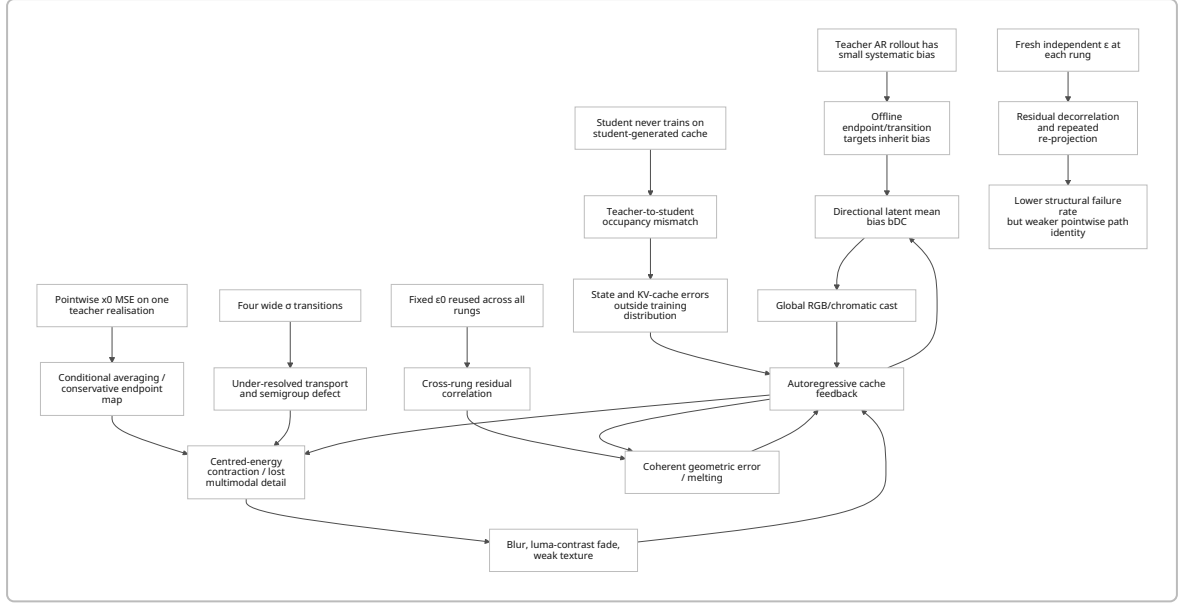
[Self-Correcting Self-Consuming Loops](#) is a useful non-video analogue. It shows theoretically that a correction map pushing generated samples towards the true distribution can make recursive synthetic-data training exponentially more stable, and empirically avoids collapse in human-motion generation even with a synthetic-to-real ratio as high as 100%. ¹⁸

For your pipeline, the corresponding correction map could be a real-video-conditioned teacher, a learned real-versus-rollout density-ratio correction, or an explicit decoder-aware DC-bias correction applied to targets. Merely increasing the quantity of teacher rollouts will reduce sampling variance but will not remove a systematic teacher rollout bias.

Unified mechanistic synthesis

A three-component model

No published paper unifies your exact pattern. The following is a synthesis supported separately by the spectral-drift, few-step-composition, stochastic-correction and on-policy-distillation literature.



At chunk level, write the latent as $z_n = \mu_n + \tilde{z}_n$. A minimal recurrence is

$$\mu_{n+1} - \mu_\star = A_\mu(\mu_n - \mu_\star) + b_{\text{target}} + b_{\text{policy}} + \eta_n,$$

$$\Sigma_{n+1} = A_\Sigma \Sigma_n A_\Sigma^\top + Q_{\text{ladder}},$$

$$g_{n+1} = A_g g_n + B_g \delta h_n + r_n,$$

where g_n denotes structural/geometric error.

The terms have distinct empirical signatures:

Component	Primary cause	Observable signature	Intervention predicted to work
b_{target} , directional mean bias	Synthetic teacher bias or biased endpoint objective	Consistent mean-vector direction; asymmetric RGB decay; survives more stochasticity	Mean-only/DC-subspace correction; real-data target anchor; bias-corrected targets
A_Σ , centred-energy contraction	Wide few-step transitions, under-resolved flow, conditional averaging	Falling latent std/ covariance eigenvalues, luma contrast and texture; strong NFE dose response	More steps; transition/ velocity loss; mixed-NFE training; covariance/low-frequency energy regularisation
Correlated r_n	Fixed noise and state-dependent endpoint error	Rung residuals have high cosine alignment; leading geometric mode grows coherently	Fresh or partially correlated re-noising; early-rung restart; semigroup regularisation
b_{policy} , $B_g \delta h_n$	Teacher-history training but student-history deployment	Failure grows with chunk index and cache replacement; weak one-chunk effect	Student-on-policy states; real/student mixed cache; self-resampled histories

How this accounts for your observations

The model predicts all of the reported dissociations:

1. **Base weights, four steps:** A_Σ is too contractive, so variance, contrast and detail fall, but b_{target} is small; the degradation is approximately hue-neutral.
2. **Base weights, 20–48 steps:** the numerical approximation is closer to the dense teacher transport, so A_Σ approaches a less contractive map. Your $0.917 \rightarrow 0.983/0.987$ retention change is the expected dose response.
3. **Fine-tuned student, four steps:** training partly compensates scalar variance contraction, explaining its better 0.956 retention, but introduces an anisotropic b_{target} , yielding a cast despite retaining more total energy.
4. **AdaIN correction:** resetting μ_n and diagonal variance suppresses both directional DC drift and some contraction; a scalar gain can only alter radial magnitude and therefore fails.
5. **Fixed-noise sampler:** correlated rung residuals excite a structural mode of the composed endpoint/cache Jacobian, causing melting even while preserving pointwise action correspondence.
6. **Fresh-noise sampler:** re-projection reduces residual coherence and therefore melting, but cannot cancel a non-zero systematic chromatic bias.

The conclusion is not that the three failures share one microscopic cause. Their unity lies in a **shared autoregressive amplifier acting on three different error subspaces**: DC mean, centred covariance and structural residual correlation.

Remedies and decisive experiments

Highest-priority experimental matrix

The most informative experiment is a factorial sweep rather than additional isolated fixes:

$$\text{weights} \times \text{NFE} \times \text{noise correlation} \times \text{history policy}.$$

Use:

- weights: base, baseline endpoint student, transition-target student;
- NFE: 4, 8, 20, 48;
- rung noise correlation:

$$\epsilon_k = \rho \epsilon_0 + \sqrt{1 - \rho^2} \eta_k, \quad \rho \in \{0, 0.25, 0.5, 0.75, 1\};$$

- context: teacher cache, student cache, mixed real/student cache.

The expected variance decomposition is:

weight identity $\rightarrow$ mean-drift direction,
 NFE $\rightarrow$ centred-energy retention,
 ρ $\rightarrow$ structural-error tail and action fidelity,
 history policy $\rightarrow$ chunk-wise amplification rate.

A strong three-way interaction between student weights, ρ and late chunk index would indicate that melting is partly an off-policy learned-field issue rather than purely a sampler property.

Separate mean from variance causally

Apply the following at chunk boundaries and, separately, after every rung:

Correction	Test	Expected result
Mean-only per channel	Replace μ_c , preserve centred latent	Removes most hue cast if DC translation is causal; limited sharpness recovery
Std-only per channel	Replace s_c , preserve mean	Restores contrast/texture more than hue
Full AdaIN	Replace both	Removes cast and some fade, reproducing current finding
Low-pass mean-only	Correct only spatial/temporal DC and very low frequencies	Similar colour correction with less interference to motion/detail
Chroma-subspace only	Correct two decoder-visible chromatic directions	Removes cast while preserving legitimate exposure/illumination evolution
Orthogonal complement only	Correct non-colour mean directions	Little RGB effect; useful negative control

Run these against both the base and student. If mean-only correction fixes the student cast but has little effect on the base blur, the training-versus-sampling dissociation becomes causal rather than correlational.

Identify the Wan VAE's decoder-visible colour subspace

For each latent channel c , add a small constant perturbation δe_c across the three latent frames and spatial field, decode, and measure the mean RGB response:

$$J_{:,c} \approx \frac{\bar{D}(z + \delta e_c) - \bar{D}(z - \delta e_c)}{2\delta}.$$

Average J over diverse real and generated latents, then take its singular-value decomposition:

$$J = U\Lambda V^\top.$$

The columns of V with non-negligible singular values give decoder-visible DC directions. Transform U from RGB into a luminance/chrominance basis, separating:

- one approximate luminance/exposure direction;

- two chromatic directions;
- a 13-dimensional approximate decoder-DC nullspace.

Project the observed chunk-to-chunk mean changes onto these directions. The directional-colour hypothesis predicts that two or three components explain most of the measured RGB mean shift even if they explain only a modest fraction of total 16-dimensional mean motion. This procedure operationalises the geometric findings of the Latent Color Subspace paper without assuming that FLUX's axes transfer to Wan. ¹⁹

Measure residual correlation rather than only output quality

For every rung, compare the student transition with the stored teacher next state:

$$r_k = x_{\sigma_{k+1}}^S - x_{\sigma_{k+1}}^T.$$

Track:

$$\cos(r_k, r_{k+1}), \quad \text{Cov}(r_k, r_{k+1}), \quad \|P_{\text{low}} r_k\|, \quad \|P_{\text{structure}} r_k\|.$$

Estimate $P_{\text{structure}}$ using leading principal components of melt-minus-non-melt residuals, DINO features, or decoder Jacobian modes. The proposed mechanism predicts:

- fixed noise: higher adjacent-rung cosine similarity and a larger leading residual covariance eigenvalue;
- fresh noise: lower cross-rung correlation but comparable or greater instantaneous residual norm;
- melting: governed more strongly by cumulative aligned residual than by mean per-rung MSE.

That distinction would explain why deterministic sampling can score better pointwise yet fail perceptually.

Test when stochasticity is useful

Use rung-specific noise correlation:

$$(\rho_2, \rho_3, \rho_4) \in \{(0, 0, 0), (0, 0, 1), (0, 0.5, 1), (0.5, 0.75, 1), (1, 1, 1)\}.$$

The ODE/SDE and ping-pong literature predicts that stochastic correction should be most valuable early or mid-path, while low- σ deterministic refinement should preserve detail and action correspondence.

²⁰

Also test **multiple fresh perturbations with endpoint averaging only during evaluation**:

$$\bar{x}_0 = \frac{1}{M} \sum_{m=1}^M F_k(x_{\sigma_k}^{(m)}, h).$$

If actual averaging greatly reduces colour drift, the bias has a substantial noise-conditional component. If it reduces melt but not mean colour, that supports the systematic-DC versus correlated-structure decomposition.

Replace final-endpoint MSE with better-aligned transport objectives

Compare:

$$\begin{aligned}\mathcal{L}_{\text{final}} &= \|F_k(x_k) - x_0^T\|^2, \\ \mathcal{L}_{\text{next}} &= \|\Phi_k(F_k, x_k) - x_{k+1}^T\|^2, \\ \mathcal{L}_{\text{disp}} &= \left\| \frac{x_{k+1}^S - x_k}{\sigma_{k+1} - \sigma_k} - \frac{x_{k+1}^T - x_k}{\sigma_{k+1} - \sigma_k} \right\|^2,\end{aligned}$$

plus a transition-distribution or adversarial/perceptual term.

Expected outcomes:

- final- x_0 MSE has the strongest conditional averaging and high- σ dominance;
- direct next-state or normalised-displacement matching improves NFE scaling and reduces blur;
- a distributional term restores sharp multimodal detail;
- a Salt-style semigroup loss

$$\|\Phi_{k \rightarrow k+2}(x_k) - \Phi_{k+1 \rightarrow k+2}(\Phi_{k \rightarrow k+1}(x_k))\|$$

reduces composed drift and fixed-noise melting. Salt's results show that this term should be combined with mixed-step/cache training rather than added in isolation. ⁵

[AnyFlow](#) additionally suggests sampling arbitrary (σ_s, σ_r) pairs rather than training only the four deployment intervals. This should make the map locally better conditioned and restore positive test-time scaling when more steps are provided. ¹⁶

Correct generated-data bias at its source

A practical progression, ordered by evidential support, is:

1. **Student-on-policy histories:** roll out the exact student sampler and train at student-visited caches, as in OPSD-V and AFD. ²¹
2. **Teacher at student states:** obtain teacher next-state or velocity targets from the student's x_{σ_k} , rather than replaying only stored teacher states.
3. **Privileged real context:** replace older generated history with encoded real frames for teacher queries, while retaining a recent student chunk to teach correction.
4. **Mixed cache quality:** alternate dense-step, four-step and deliberately degraded histories, as in Salt and Self-Resampling. ²²
5. **Real-data anchor loss:** retain a non-zero fraction of real continuations or teacher evaluations under real history. Synthetic-loop work consistently finds that access to the original distribution or a correction towards it is protective. ²³
6. **Bias-corrected teacher targets:** estimate the teacher's mean DC drift conditioned on chunk index, action and scene statistics, then remove only the decoder-visible chromatic component before student regression.
7. **Student-history DAGger schedule:** begin with teacher histories, progressively replace chunks with student generations, and query a dense teacher for corrections at the resulting states.

A useful audit is to estimate the teacher and student mean-drift operators:

$$\Delta\mu_n^T = \mu_{n+1}^T - \mu_n^T, \quad \Delta\mu_n^S = \mu_{n+1}^S - \mu_n^S.$$

Regress $\Delta\mu_n^S$ on $\Delta\mu_n^T$. An amplification coefficient greater than one along the leading chromatic direction would directly demonstrate inherited teacher bias followed by student/AR amplification.

Avoid over-correcting legitimate colour evolution

Pinning all 16 means and standard deviations to the initial real seed can suppress genuine illumination change, fire/light effects, camera exposure changes or action-dependent scene evolution. The safer hierarchy is:

- anchor only decoder-visible chroma, not luminance;
- use an EMA or slowly varying expected statistic rather than a fixed seed;
- condition the reference on action, text and predicted lighting change;
- apply correction only when drift exceeds a real-video-calibrated confidence envelope;
- regularise mean increments rather than forcing exact equality:

$$\mathcal{L}_{DC} = \|P_{\text{chroma}}(\Delta\mu - \Delta\mu_{\text{ref}})\|^2.$$

FreqForcing’s repetition trade-off under strong spectral anchoring and Self-Resampling’s content drift under overly strong corruption both show why correction strength must be calibrated rather than maximised. ²⁴

Ranked literature

Most relevant papers

1. [Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion](#) — the closest overall system match: Wan2.1, causal 3D VAE, three-latent-frame chunks, four-step shifted ladder, fresh-noise endpoint re-noising and cached AR context; its main difference is distribution-level self-rollout training rather than stored teacher-transition MSE. ¹
2. [FreqForcing: Autoregressive Long Video Generation via Spectral Self-Anchoring](#) — the strongest direct video diagnosis of colour drift, fading and DC/low-frequency rollout energy drift, with quantitative long-video correction results. ²⁵
3. [Stable Audio 3](#) — the exact procedural match to fresh-noise x_0 -predict/re-noise-to-lower-absolute-level sampling, and the clearest empirical argument that this operation can correct deterministic trajectory displacement. ⁹
4. [OPSD-V: On-Policy Self-Distillation for Post-Training Few-Step Autoregressive Video Generators](#) — the closest direct remedy for teacher-cache versus student-cache distribution shift in a four-step, three-latent-frame Wan AR model. ²⁶
5. [Salt: Self-Consistent Distribution Matching with Cache-Aware Training for Fast Video Generation](#) — the strongest published link between low-NFE over-smoothing, incoherent composition of endpoint updates, cache-quality variation and semigroup-style correction. ⁵
6. [Why Do DiT Editors Drift? Plug-and-Play Low Frequency Alignment in VAE Latent Space](#) — the closest empirical precedent for correcting repeated-transition drift by low-pass per-channel mean and variance alignment rather than scalar rescaling. ²⁷

7. [The Latent Color Subspace: Emergent Order in High-Dimensional Chaos](#) — the strongest evidence that decoded colour corresponds to low-dimensional directions in globally averaged VAE latent space, enabling decoder-aware subspace correction. ¹⁹
8. [AnyFlow: Any-Step Video Diffusion Model with On-Policy Flow Map Distillation](#) — the closest alternative formulation to your progressive transition target, replacing fixed endpoint consistency with arbitrary-time, on-policy flow-map learning that retains positive step-count scaling. ¹⁶
9. [End-to-End Training for Autoregressive Video Diffusion via Self-Resampling](#) — direct quantitative evidence that histories containing structured, autoregressively generated model errors are better training exposure than clean teacher forcing or generic Gaussian corruption. ⁴
10. [Restart Sampling for Improving Generative Processes](#) — the strongest established theory that stochastic noise injection followed by learned transport can contract accumulated model/solver error, although its upward restart cycles differ materially from your monotone ping-pong ladder. ¹¹

Overall conclusion

The combination you observe has **no single published precedent**. The closest literature-wide explanation is nevertheless coherent and testable:

- **training** can introduce an anisotropic endpoint bias in the decoder-visible latent DC subspace, producing compounding colour cast;
- **few-step sampling** can contract centred latent covariance and low-frequency energy, producing hue-neutral blur and contrast fade even with unchanged weights;
- **fixed noise reuse** can preserve a coherent target path but also preserve the sign and geometry of learned endpoint errors, allowing a structural mode to grow through rung composition and AR cache feedback;
- **fresh-noise endpoint re-projection** reduces that cross-rung coherence and repeatedly returns the state to a neighbourhood the endpoint predictor recognises, improving structural robustness without cancelling systematic mean bias;
- **teacher-generated, teacher-forced rollout training** transmits teacher bias and leaves the student untrained on its own cache distribution, raising the autoregressive gain of all three failures.

The strongest novel contribution in your evidence is not any one symptom, but the controlled factorisation supplied by the base/student, 4/20/48-step, scalar/AdaIN and fixed/fresh-noise ablations. A decoder-Jacobian colour-subspace measurement, mean-only versus variance-only correction, and a continuous noise-correlation sweep would turn that factorisation into a substantially stronger mechanistic result than currently exists in the literature.

¹ <https://arxiv.org/abs/2506.08009>
<https://arxiv.org/abs/2506.08009>

² ²⁴ ²⁵ <https://arxiv.org/html/2607.27110>
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